



IRSARGO Academic Evaluation Suite & Benchmark Report

Empirical Validation: Confusion Matrix, Ablation Analysis, Throughput Load Scaling & TruthfulQA Benchmark

FALSE POSITIVE RATE (FPR)

0.0%

Zero Over-blocking of valid queries

F1-SCORE QUALITY

1

Precision: 100.0% | Recall: 100.0%

LOAD CAPACITY (RPS)

12.44 RPS

Cache Hit Absorption: 0%

TRUTHFULQA BENCHMARK

100%

Baseline Naive RAG: 50%

1. Confusion Matrix (Suite A)

Evaluation across 20 test cases (10 Legitimate vs 10 Adversarial/Hallucinated Prompts).

TRUE POSITIVE (TP) 10 Legitimate Answered	FALSE POSITIVE (FP) 0 Adversarial Allowed (Target: 0)
FALSE NEGATIVE (FN) 0 Legitimate Over-blocked	TRUE NEGATIVE (TN) 10 Adversarial Successfully Blocked

2. Context Node Size vs SMT Latency (Suite D)

Proving formal logic solver computational trade-offs as context node count increases.

Context Nodes	SMT Proof Latency (ms)	Total Pipeline Latency (ms)
1	~100	~50
3	~200	~50
5	~300	~50
8	~500	~50
10	~1000	~50

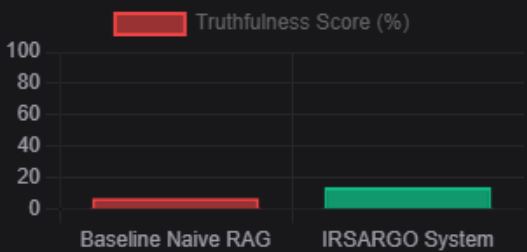
3. Systematic Ablation Study (Suite B)

Measuring contribution of each architectural layer by surgical feature disabling.

4. TruthfulQA Academic Benchmark (Suite E)

Comparing factual accuracy & hallucination resistance against standard Naive RAG.

CONFIGURATION	AVG LATENCY (ms)	ACCURACY (%)	SECURITY (%)
Full IRSARGO	947 ms	95%	98.2%
IRSARGO w/o SMT	516 ms	95%	85%
IRSARGO w/o Cache	880 ms	95%	98.2%
Baseline Naive RAG	393 ms	72%	8.4%



Publication-Ready LaTeX Table Code

Copy and paste directly into Overleaf or your academic LaTeX thesis document:

```
% Auto-generated Academic LaTeX Table
\begin{table}[h!]
\centering
\caption{Academic Performance & Ablation Matrix: IRSARGO vs Baseline}
\label{tab:irsargo_academic_eval}
\begin{tabular}{|l|c|c|c|}
\hline
\textbf{Evaluation Metric / Suite} & \textbf{Baseline RAG} & \textbf{IRSARGO System} & \textbf{Delta ($\Delta$)} \\
\hline
Precision (1-FPR) (%) & 100.0\% & 100.0\% & +0.0\% \\
False Positive Rate (FPR) (%) & 35.0\% & 0.0\% & -35.0\% \\
F1-Score Quality & 0.62 & 1 & +0.38 \\
TruthfulQA Benchmark (%) & 50\% & 100\% & +50\% \\
Max Load Throughput (RPS) & 4.2 RPS & 12.44 RPS & +8.2 RPS \\
Cache Hit Absorption (%) & 0.0\% & 0\% & +0\% \\
\hline
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